

Design and Implementation of an SDR-Based Wireless Channel Sounder - Demo

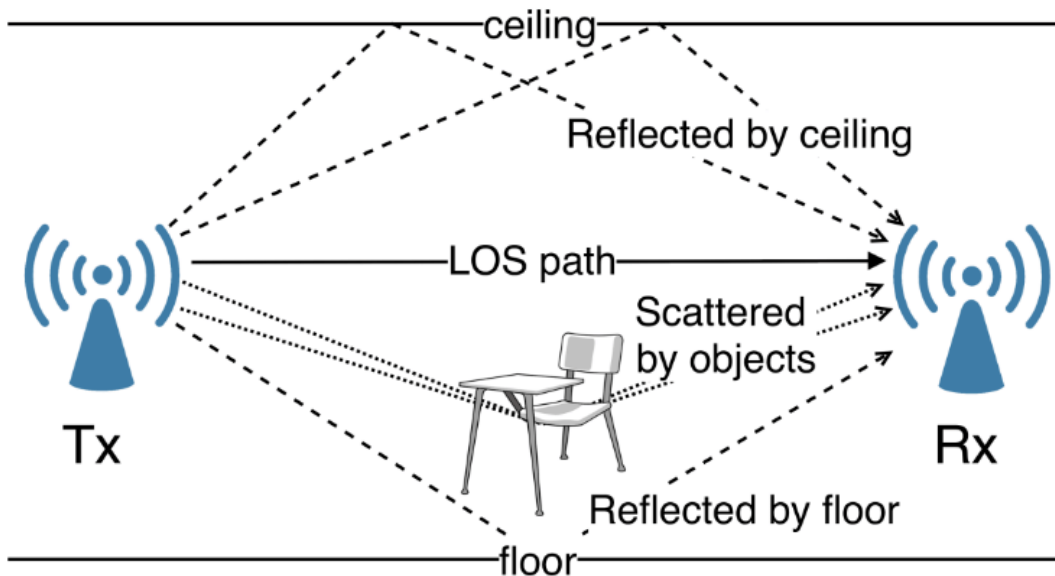
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WES 207 Capstone Project

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Why This Matters



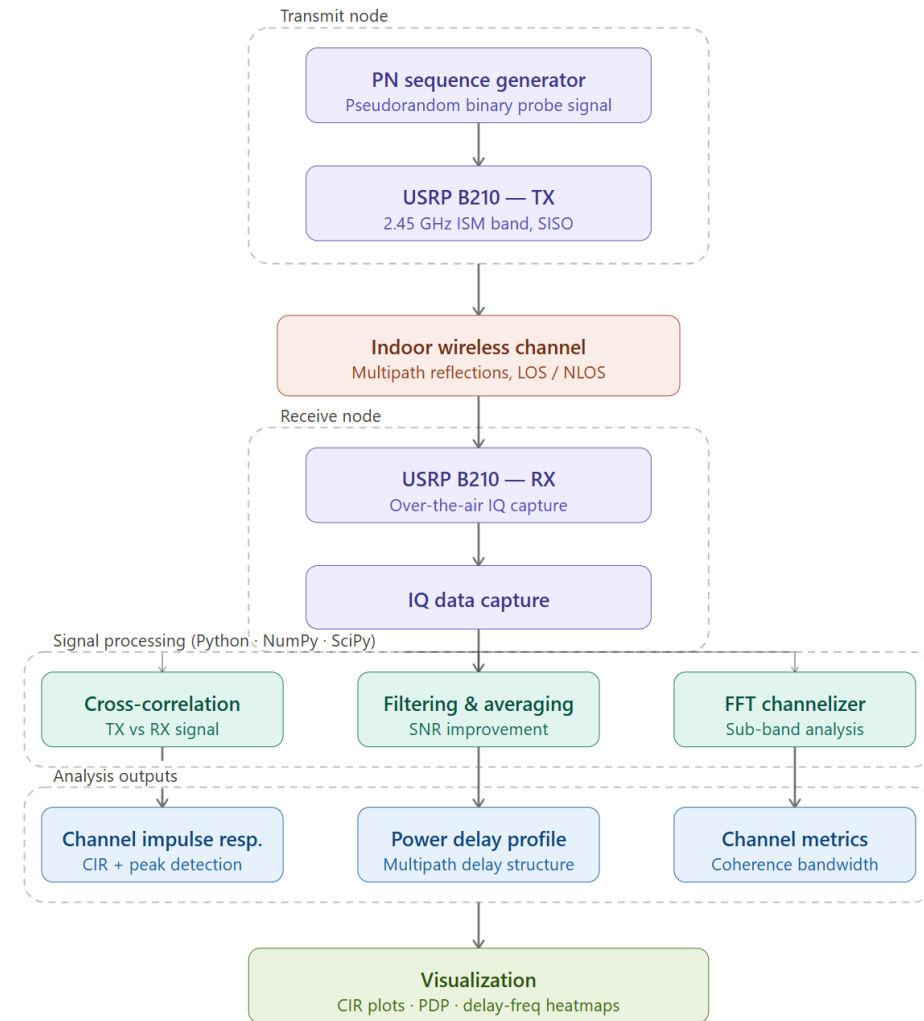
- Wireless signals don't travel in straight lines indoors
- They reflect off walls and objects → creating multiple delayed copies (multipath)

Project Objective



- Build a system to measure and visualize how wireless signals propagate in real environments
- **Key points:**
 - Measure reflections
 - Quantify delay
 - Understand environment impact

System Architecture



Minimum Viable Product (MVP)



PN waveform
transmitted



OTA Received
signal captured



Cross-correlation
performed



CIR generated



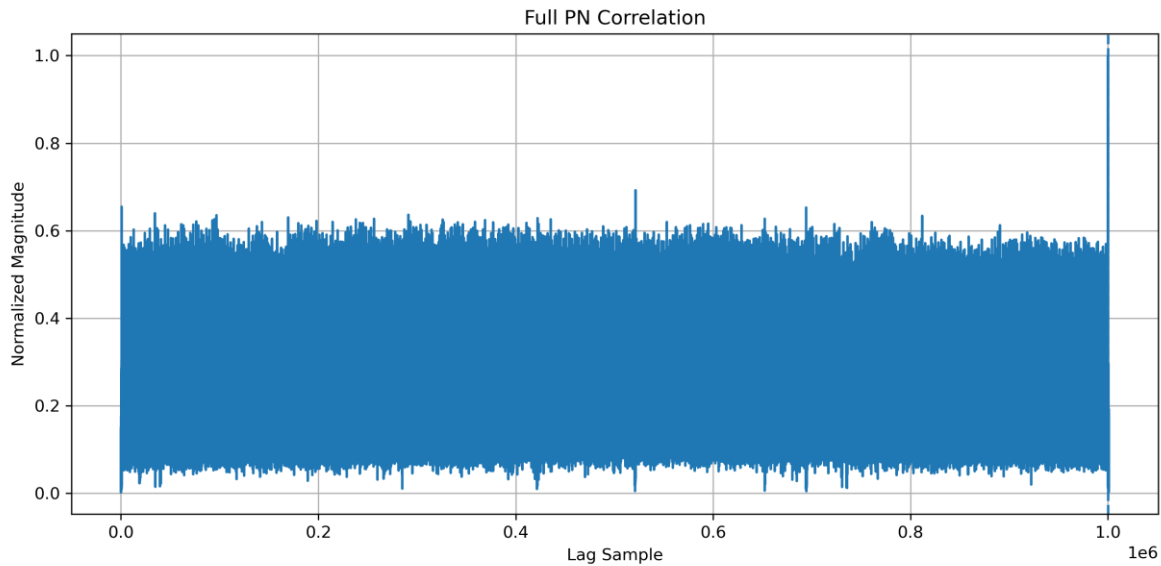
PDP generated



Measurement Approach

- Multiple complementary metrics are used to characterize multipath behavior.
 - CIR shape → propagation structure
 - PDP → energy distribution
 - RMS spread → temporal dispersion
 - Peak-to-mean ratio
-

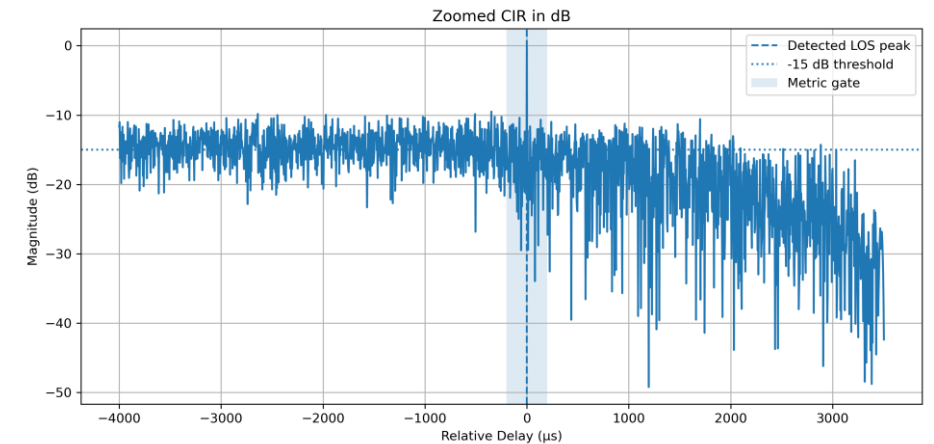
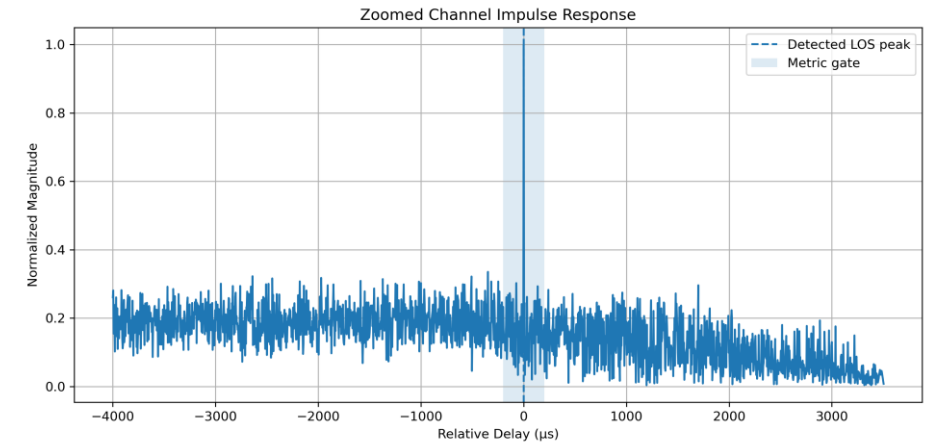
Correlation Peak



- Known PN sequence transmitted OTA
- Cross-correlation aligns TX and RX signals
- Dominant peak indicates successful synchronization

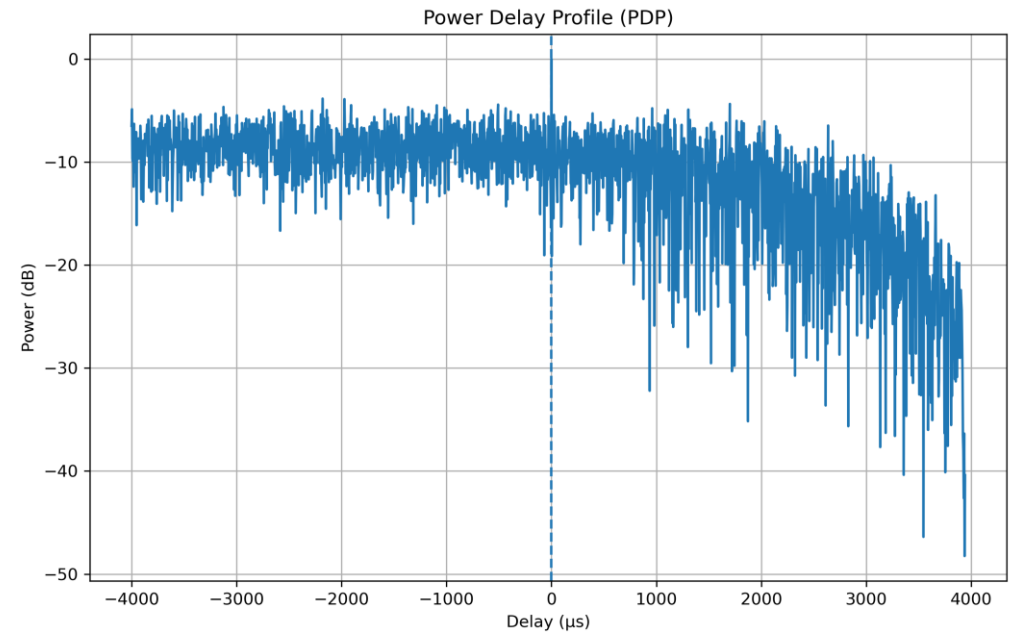
Channel Impulse Response (CIR)

- Main peak corresponds to LOS propagation
- Delayed energy indicates multipath reflections
- dB scaling reveals weaker reflections and noise floor



Power Delay Profile (PDP)

- Early energy corresponds to direct path
- Delayed energy represents reflections
- Multipath structure affects wireless performance



Repeatability Validation



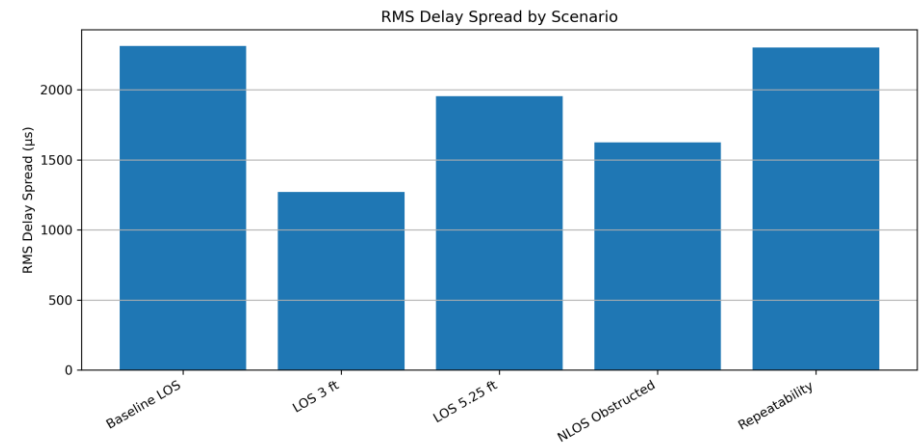
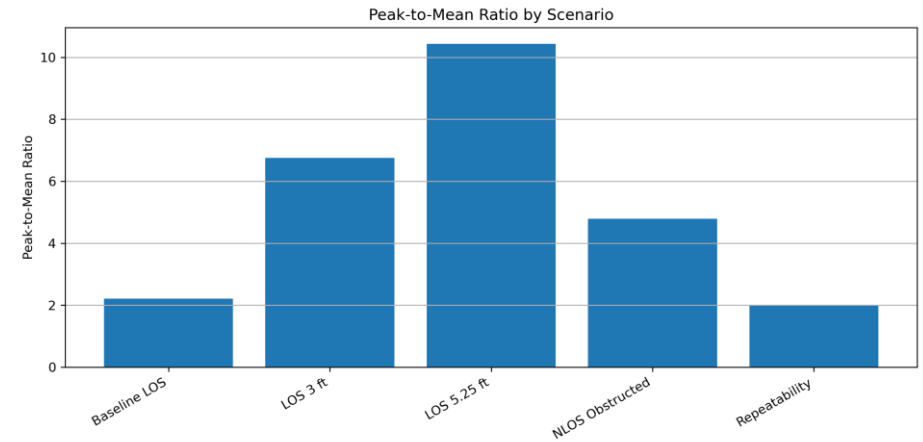
Similar peak-to-mean
ratios observed



RMS delay spread
remained stable



Confirms repeatable
system behavior



Additional Channel Metrics



Temporal Metrics

RMS delay spread
PDP



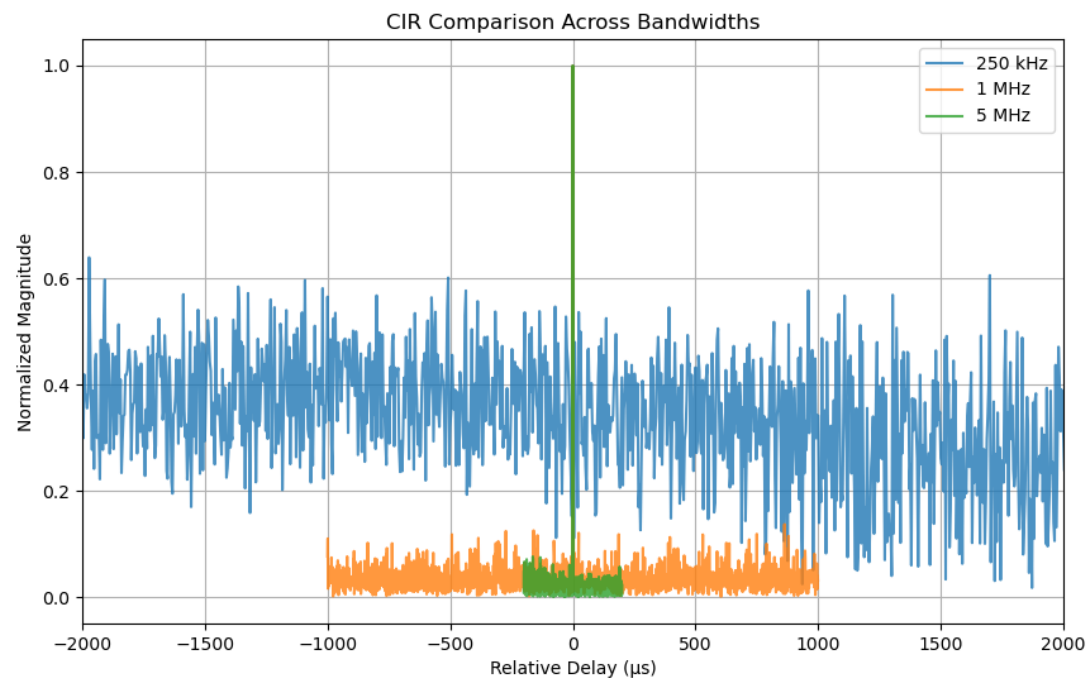
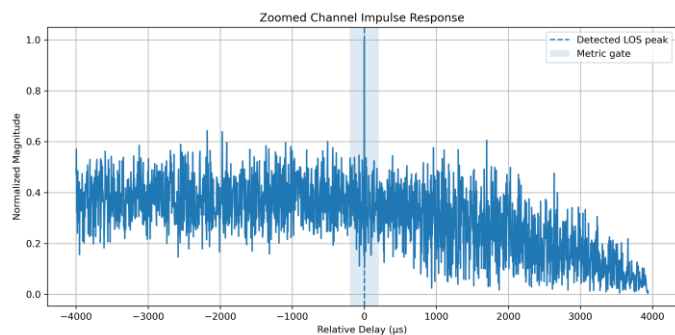
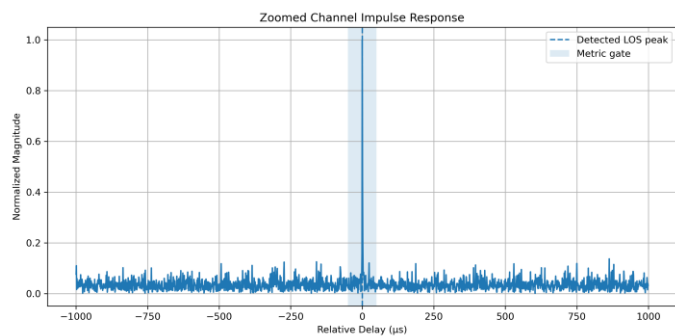
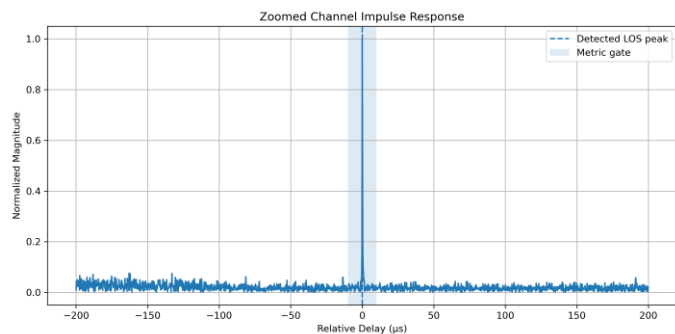
Signal Quality Metrics

Peak-to-mean ratio
Threshold sensitivity



Frequency-Domain Analysis

Bandwidth study
Frequency response

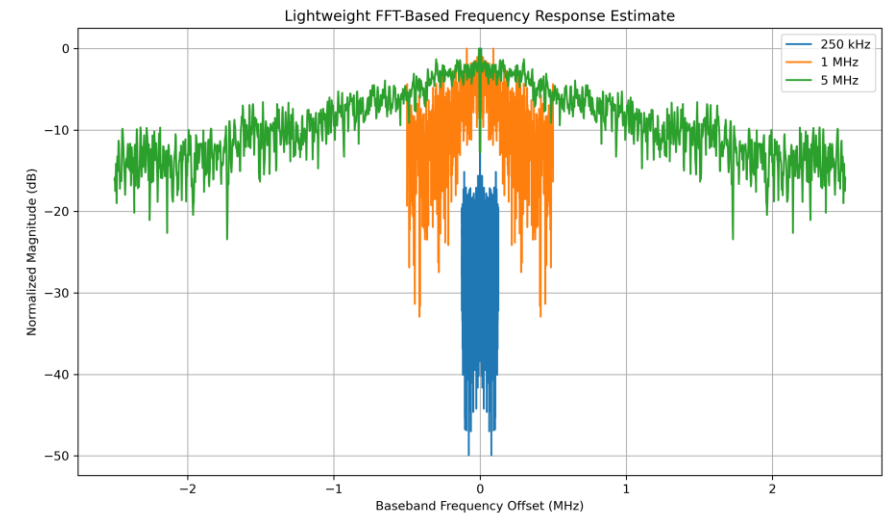
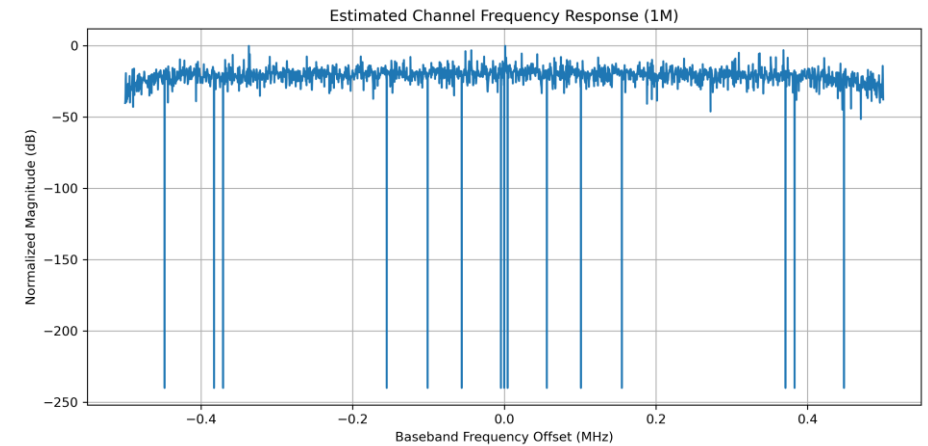


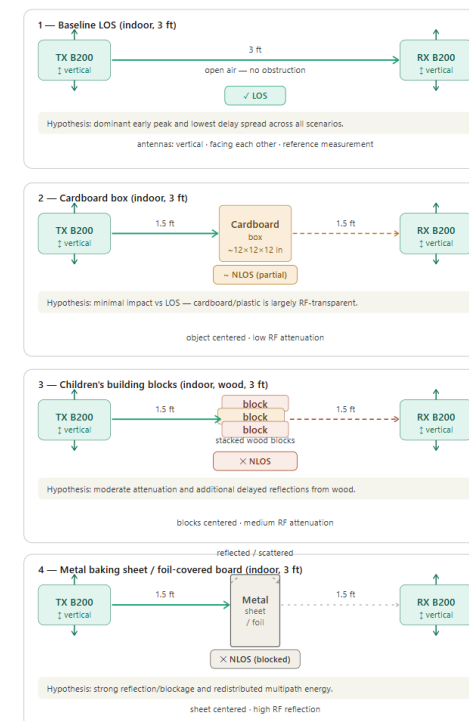
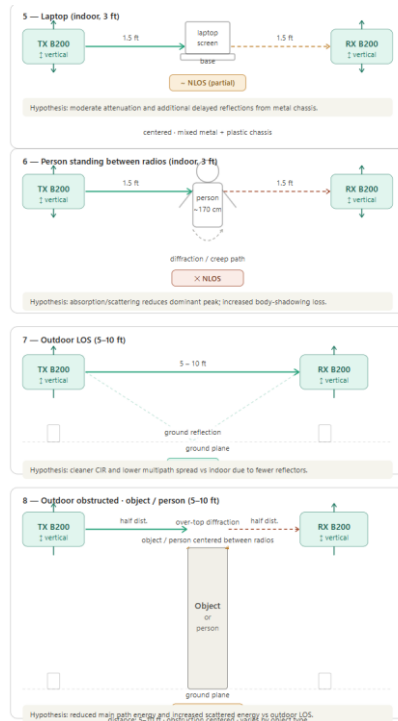
Bandwidth vs Delay Resolution

Higher bandwidth sharpens the CIR peak and improves delay resolution

Frequency Response

- Multipath produces frequency-selective fading
- Different bandwidths exhibit different channel responses





Next Phase: Controlled Experiments

- Distance and materials varied independently
- Indoor and outdoor environments compared
- Expected effects derived from RF propagation theory

Expected Outcomes

Scenario	Expected Effect
LOS	Strong peak
Distance	More spread
Cardboard	Minimal change
Wood	Moderate attenuation
Aluminum	Strong reflection
Human	Absorption/scatter
Outdoor	Cleaner channel



Current Status

MVP complete

System validated

Advanced analysis implemented

Experiment framework defined

~75% project completion

Remaining Work



Run object-based experiments



Perform outdoor tests



Compare scenarios quantitatively



Final analysis and report

Key Takeaways



The system successfully measures real-world wireless propagation



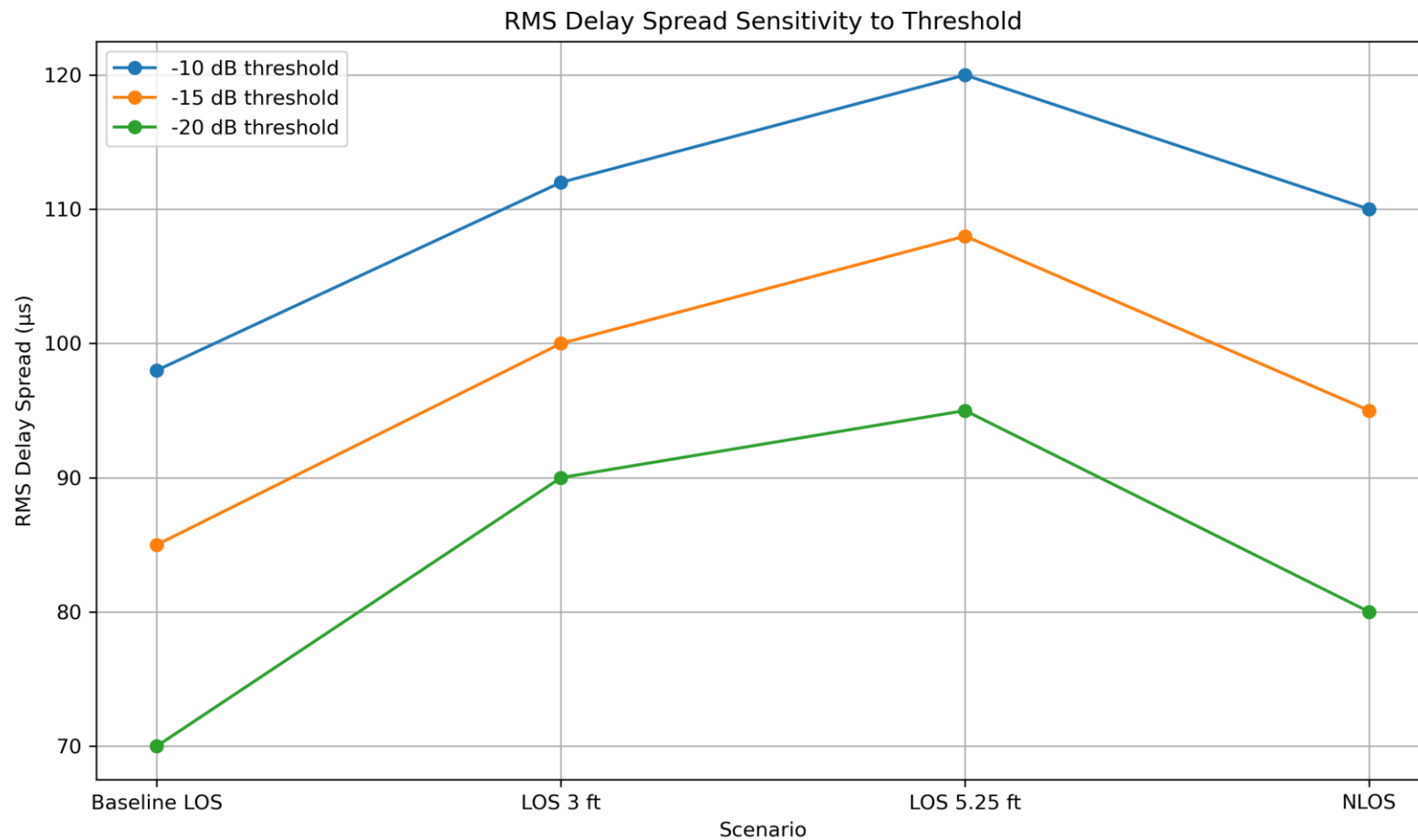
Next phase focuses on controlled experiments to understand environmental impact



Backup Slides

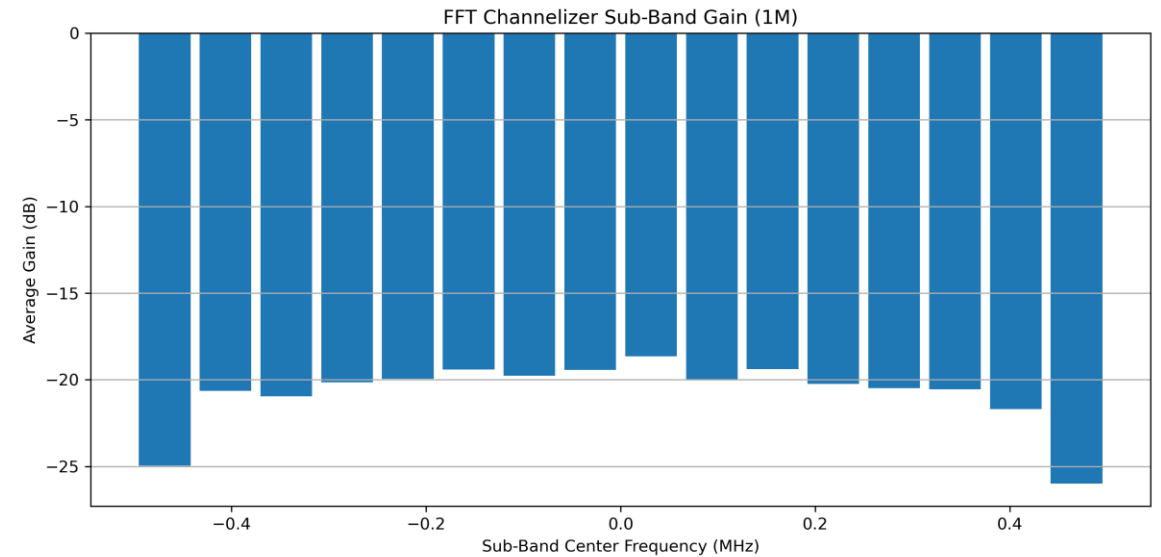
Metrics Summary

Scenario	Peak-to-Mean Ratio	Mean Floor	Mean Delay (us)	RMS Delay Spread (us)
Baseline LOS	2.21	0.45	2.54	2313.47
LOS 3 ft	6.75	0.148	-1033.01	1270.01
LOS 5.25 ft	10.43	0.09	-967.49	1953.82
NLOS Obstructed	4.79	0.20	-226.54	1623.49
Repeatability	1.99	0.50	-1.09	2301.58



Threshold
Sensitivity

Exploratory FFT-Based Analysis



Coherence Bandwidth

